

**Solution:** From Theorem 3, we know that the given statement is True.

b) Any linear combination of vectors can always be written in the form  $Ax$  for a suitable matrix  $A$  and vector  $x$ .

**Solution:** On Page 36 just above Theorem 3, it is mentioned that any vector equation can be written as an equivalent matrix equation in the form  $Ax=b$ . For example,  $x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$  is equivalent to

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}. \therefore \text{The given statement is True. } \square$$

c) The solution set of a linear system whose augmented matrix is  $[a_1 \ a_2 \ a_3 \ b]$  is the same as the solution set of  $Ax=b$ , if  $A = [a_1 \ a_2 \ a_3]$ .

**Solution:** From Theorem 3 on Page 36, the given statement is True.  $\square$

d) If the equation  $Ax=b$  is inconsistent, then  $b$  is not in the set spanned by the columns of  $A$ .

**Solution:** From the topmost grey box of Page 37 we know that the equation  $Ax=b$  has a solution if and only if  $b$  is a linear combination of the columns of  $A$ .  $\therefore$  The equation  $Ax=b$  does not have a solution if and only if  $b$  is not a linear combination of the columns of  $A$ . From the defn. of span,  $b$  is in the set spanned by the columns of  $A$  if and only if  $b$  is a linear combination of the columns of  $A$ .  $\therefore$  The equation  $Ax=b$  does not have a solution if and only if  $b$  is not in the set spanned by the columns of  $A$ . From the defn. of inconsistency, if the equation  $Ax=b$  is inconsistent, then  $b$  is not in the set spanned by the columns of  $A$ .  $\therefore$  The given statement is True.  $\square$

e) If the augmented matrix  $[A \ b]$  has a pivot position in every row, then the equation  $Ax=b$  is inconsistent.

**Solution:** From Ex-23(c), we know that the given statement is False.  $\square$

f) If  $A$  is an  $m \times n$  matrix whose columns do not span  $\mathbb{R}^m$ , then the equation  $Ax=b$  is inconsistent for some  $b$  in  $\mathbb{R}^m$ .